ANALOG ALGORITHMS: GENERATIVE COMPOSITION IN MODULAR SYNTHESIS

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ABSTRACT

The contemporary re-emergence of modular synthesisers as a popular tool for music making rejects much of the conveniences afforded by advancements in music technology in the past 40 years, an idea that challenges our understanding of the relationship of electronic music composers to technology. Given the dominance of the computer in music making since the early 1990s there is a temptation to situate the modular synthesiser in opposition, focusing upon the modular instrument's tangible interface and a general shift away from the ubiquitous computer as primary drivers for the return to modular synthesis (Paradiso 2017). I argue that generative compositional approaches, specific to modular synthesisers, are another factor that should be considered in relation to this re-emergence.

The modular synthesis paradigm foregrounds generative compositional processes designed not only to create sound, but to generate musical events and control signals. Many module designs carry a legacy of generative processes that can be traced back to the earliest commercial synthesisers. These generative approaches exhibit attributes that are intrinsically distinguishable from those developed in the field of computer music.

1. INTRODUCTION

Modular synthesisers have recently re-emerged as a popular tool for electronic music composition (Roads 2015, 88). Paradiso argues that the causes attributed to this renewed interest are driven by a desire for tangible interfaces, a reaction to the omnipresence and continual obsolescence of computers and software, the collecting behaviour of consumers, and the democratisation of the design and manufacturing processes used (Paradiso 2017). This paper argues that another, largely overlooked, factor in the contemporary resurgence of modular synthesisers is the generative compositional strategies they afford.

In the following section, a historical context is provided emphasising a shift in the popular understanding of what a synthesiser does or is designed to do. An analysis then follows of the historic discourse associated with several significant instruments, revealing a paradigm of algorithmic electronic composition that has developed since the 1960s (Bode, 1961). Several generalisable experiential attributes identified through my own reflective practice of using modular synthesisers are then described. This serves firstly toward understanding the possible impact of these generative compositional processes, and secondly, toward a broader understanding of the recent resurgence in the use of modular synthesisers.

2. HISTORICAL CONTEXT

It is generally accepted (Manning 2013) that Harald Bode designed and built the first modular synthesiser in 1961. Bode credits analog computing as the inspiration for the modular approach, enabling the creation of "intricate electronic systems from modules with a limited number of basic functions" (Bode 1961, 5). Donald Buchla and Robert Moog then went on to develop and commercially release their own modular synthesiser[1] designs in 1966 (Manning 2013, 102).

Up until the early 1970s these large, expensive instruments experienced a level of commercial success (Vail 2000, 41), in particular, Moog's company R. A. Moog Inc benefited from the attention generated by the hit record *Switched on Bach* by Wendy Carlos (Vail 2000, 41). Buchla was less concerned with commercial success (Pinch and Trocco 2004, 33), coming from the perspective of a tape music composer with an interest in John Cage and indeterminism (Goebel & Rockwell, 2008, p. 173) and embedded in the counterculture movement (Pinch and

[1] The term 'synthesiser' is deliberately used throughout this article in its contemporary usage encompassing a range of instruments and methods. The term was used by Harald Bode in his article *Sound Synthesizer Creates New Musical Effects* (Bode 1961). Donald Buchla used a range of other terms including 'Buchla Box' (Pinch and Trocco 2004) and 'Modular Electronic Music System' (Buchla 1966) and there was likely an intention behind this terminology, however, the term has since stabilised to describe most electronic instruments that formulate audible waveforms.

Trocco 2004, 35). Whilst Moog's modular synthesiser sales dropped off sharply in 1970 (Vail 2000, 41), a new instrument design, the portable Minimoog D, was building momentum and would eventually be sold in the tens of thousands (Holmes 2015, 268).

The Minimoog D had a number of advantages over the larger modular instruments; it had a fixed architecture design, with standardised connections between components, greatly simplifying its design. The inclusion of an integrated keyboard interface made its function more apparent to potential users (Paradiso 2017, 2), particularly as a performance instrument. The instruments smaller size allowed it to be portable and its cost made it more accessible. The success of the Minimoog drove other companies to develop similar fixed architecture, keyboard based instruments such as the ARP Odyssey (Pinch and Trocco 2004, 265). These new, compact designs forgoed much of the flexibility of the proceeding modular designs (Holmes 2015, 248) and firmly established the keyboard as the standard interface for synthesisers (Pinch and Trocco 2004, 8).

Jumping forward to 1983, the MIDI standard was established (Manning 2013, 267), the *note on, note off*, paradigm drawn directly from the piano keyboard (Dalgleish 2016). The ubiquitous and relatively consistent nature of the MIDI protocol paved the way for a more complete separation of interface and instrument, enabling sound modules and rackmount synthesisers to be developed that were entirely dependent on external MIDI control or sequencing devices in order for sound to be emitted.

The preceding compressed historical narrative highlights a cementing of the definition of *synthesiser* as a device that produces sound (a sound engine), and a conceptual separation of the sound engine from devices that provide musical or other controlling data such as keyboards or sequencers.

Various synthesizer instruments have included sequencing or arpeggiator devices capable of controlling the synthesiser sound engine, some of these sequencers such as the Roland TB-303 exhibit unique attributes, and may be perceived to contribute key qualities to the instrument's character. In these cases the conceptual separation of controller and sound engine persists as the sound engine cannot impact upon the operations of the sequencing device, the relationship of control is in one direction, and the two processes of sound producing, and event and parameter control generation, remain distinct.

3. ANALOG ALGORITHMS

Timing event generators and sequencers were an integral component of the first Buchla synthesiser (Vail 2000). The design impetus came from tape music practices, as a means to speed up compositional processes through the automating of *sequences* of pitch and amplitude changes (Hanson 2010, 18). Three sequencing modules were included in the original Buchla 100 (also known as the Buchla Music Box or the Buchla Electronic Music System), two Model 123 Sequential Voltage Source modules with 8 stages, and the 16 stage Model 146 Sequential Voltage Source (Vail 2000, 107). The modules were identical aside from the number of stages offered (H. Howe 1967, 24), both offering 3 voltage level outputs per stage and a rotary switch to limit the number of stages. A timing pulse triggers the modules to select the next set of voltages, after reaching the final stage the sequencer would then return to the first stage. The Model 140 Timing Pulse Generator (H. Howe 1967, 21) was designed to deliver voltage pulses that would provide a source of triggers for the various sequencer, envelope generators and sample and hold based modules of the Buchla 100. The timing of the pulses generated could be controlled by an external voltage source and the 1967 Buchla User Manual suggests a circular relationship between the Model 140 and the sequencer modules enabling each step to be of differing length (H. Howe 1967, 22).

The inclusion of these devices, in one of the earliest modular synthesisers demonstrates the purpose and function of these instruments as more than sound generators. Instead emphasising the generation and control of musical events. Morton Subotnick described his compositional techniques using the Buchla 100, where complex rhythms could be generated over long periods of time by combining the various sequencers at odd, differing lengths, or applying voltage from other sources such as envelope generators to adjust the timing of a pulse generator (Vail, 2000, p. 109). The possibilities for the combination and complex interaction of these musical event generating devices, involves a process identical to the creative design of synthesised sounds using the modular synthesiser. It is possible that in the early stages of the emergence of synthesiser instruments, the processes of designing sounds, and generating or manipulating musical events would have been difficult and unnecessary to differentiate.

Curtis Roads offers an analysis of the time scales (Roads 2004, 19) describing 9 sections of scale from infinitesimal (short) to the infinite (long). The description of a continuous spectrum encompassing event frequencies that are understood as sound, through to musical events and structures, resonates with the possibilities encouraged by the modular synthesiser instrument. The sections of the time scale of particular relevance are Macro, Meso, Sound Object and Micro;

> Macro - *The time scale of overall musical architecture or form, measured in minutes or hours, or in extreme cases, days.*
>
> Meso - *Divisions of form. Groupings of sound objects into hierarchies of phrase structures of various sizes, measured in minutes or seconds.*
>
> Sound object - *A basic unit of musical structure, generalizing the traditional concept of note to include complex and mutating sound events on a time scale ranging from a fraction of a second to several seconds.*

Micro - *Sound particles on a time scale that extends down to the threshold of auditory perception (measured in thousandths of a second or milliseconds)*. (Roads 2004, 19)

Subotnick's description of the implementation of an envelope to modulate the timing of a pulse generator that in turn controls the speed of a sequencer effectively shifts the role of the envelope from a *sound object* shaping device, (amplitude, oscillator frequency, filter frequency etc) to the *meso* structural manipulation and production of *sound objects (triggering of sequencer events, modulating phrase tempos, triggering of envelopes etc)*. Modular synthesisers also enable a blurring of purpose and effect from the *sound object* level up to the *micro* where signals become audible. However Buchla made a deliberate design decision to separate audio signals, control voltages and timing pulses (Austin, Kahn, and Gurusinghe 2011, 301), as a result the possibilities for components to transverse from sound object event up to audible frequency or vice versa were reduced. However this is still possible using a Buchla 100 system, for example; a timing generator can operate at high frequencies and in turn trigger a sequencer to modulate an oscillator at an audible frequency to produce complex FM style synthesis.

In contrast to Buchla's approach, the Moog systems approach to control voltage did not differentiate between audio and control signals, enabling many modules to interact with one another (Dalgleish 2016), however different connections were used for gate and trigger sources preventing interconnection with other signals without the use of a converter. The design of the Moog synthesisers was more closely linked to the needs of traditional musician (Pinch and Trocco 2004, 61) and the potential for generative approaches are more limited than those offered by the Buchla equivalent, a point acknowledged by Moog:

There are also people like Morton Subotnick and Suzanne Ciani who are concerned, as Cage was, with production of music as a process, where to realize your music, you would organize a very complex system. The Buchla modular system was designed with this sort of composer in mind more than ours was. It has a lot of capability for triggering sources in sequence, for turning on and off different sources, and for creating a very complex organization of a modular system. You can literally set up a machine that will produce an interesting sounding piece of music by itself. (Hutchins 1974)

Serge Tcherepnin worked with Morton Subotnick in a New York based studio affiliated with New York University in the late 1960s. The studio was centered around a Buchla synthesiser that Subotnick brought with him from San Francisco (B. Gluck 2012, 65, 66). Tcherepnin had worked with electronics since he was a child (Vail 2000, 152) and began to develop circuitry at the New York studio (R. J. Gluck 2009) (B. Gluck 2012, 70). In 1969 Subotnick moved to California Institute of the Arts (CalArts) as Dean of the Music Department ("About Morton Subotnick" n.d.). Tcherepnin also became a Professor there in 1970 (Vail 2000, 149).

Frustrated by the high price of modular synthesiser systems Tcherepnin, worked with Randy Cohen and Richard Gold, to design and built the first 'Serge' synthesiser (initially referred to as 'Tcherepnins') in 1973 (Vail 2000, 149–50). Based on this prototype a further 20 were assembled on the CalArts campus by a coalition of staff, students and 'outside agitators', each participant contributed $700 to cover the cost of materials and after several months each received their own synthesiser (Vail 2000, 150). Tcherepnin left CalArts in 1975 to start the Serge company and continue development of his designs.

The influence of Buchla's approach can be observed in the design of the Serge modular. Specific modules such as the Serge Dual Slope Generator (DSG) are immediately reminiscent of the Buchla equivalent Model 280 (also Models 281 and 284) in that both devices are centred around simple envelopes with adjustable attack and release controls. Tcherepnin's DSG design also enabled the module to act as a voltage processing device capable of 'slewing' the rate of change of an incoming signals' direction either upwards or downwards (Gold, Johansen, and LaPalma 1976, 5–18). This process served much like a low pass filter. In addition, a trigger output was provided to report the end of the envelope's cycle. These design innovations enable the module to be used for a range of other purposes in addition to producing an envelope including; generating subharmonic frequencies from oscillators, envelope following of an external signal, delaying a trigger or gate and acting as a rudimentary audio filter. This flexible approach is typical of Tcherepnin's concept of *patch programmability* (Tcherepnin 1979, 2) where modules can perform a range of distinct functions, this approach reflects Serge's intentions to create cheaper, smaller and more flexible modular synthesisers (Vail 2000, 150).

The Serge modular system does not differentiate between control and audio signals, with the exception of audio output and inputs from outside of the synthesiser system itself (as it otherwise utilises banana plugs with a common ground). Serge oscillators were also designed to continuously sweep from 1hz to 100khz without any range switching (Vail, 2000, p. 151), enabling them to operate across time scales to produce sound object level musical events or audible waveforms. Tcherepnin describes his own interest in electronic music as being a foregrounding of the electronics themselves;

The Buchla's modules made me realise how much more interesting his modules would be, if he had made available to the user the "hidden" sub- functions making up a module. In a sense, dissecting a Buchla module, I could do the same stuff I did with a Japanese transistor radio.

This was the guiding principle of the early Serge: make available electronic functions that are interesting in themselves, though seemingly "un-musical". Thus

modules like the positive slew, negative slew, peak and trough, comparators, processors and wave shapers, etc. came into existence.

(Biddulph et al. 2018, sec. Foreword)

John Cage is quoted as remarking that the Serge synthesiser was "the best musical composition that Serge had ever made" (Holzer 2011), perhaps in jest, but nevertheless acknowledging the generative and compositional nature of these instruments. Serge synthesisers continued to be produced in limited numbers throughout the 1980s, 90s and 00s, sometimes in kit form, many module designs are still available today, manufactured under license by various companies.

The Eurorack modular standard established by Dieter Doepfer in 1995 (Dalgleish 2016) has emerged at the forefront of the more recent resurgence in modular synthesisers. According to the ModularGrid website database there are currently 342 module manufacturers (as of February 2019), and nearly 5600 modules. The legacy of historic modular synthesiser design is apparent in many modules; from licensed manufacturing of older designs to new designs that clearly build upon specific models or concepts. For example the popular Maths module by Make Noise, of which more than 5000 units had been sold by late 2016 (Grosse 2016) draws heavily upon the design of the Serge DSG (also DUSG), adding a two channel invertable mixer and logical OR output inspired by the Buchla Model 257 (Rolando 2009). Through these contemporary designs, compositional approaches and processes have endured and advanced since their origins in the 1960s and 70s.

4. QUALITIES OF MODULAR SYNTHESISER GENERATIVE COMPOSITION

The specific possibilities for composition using a modular synthesiser are endless, the expandable and completely customisable nature of the instrument itself compounds this. However it is still possible to identify some overarching factors that can be generalised to impact upon generative compositional processes in modular synthesis. The following attributes have been subjectively developed through a process of reflection upon my own compositional experiences with modular synthesisers (Schön 1983).

4.1. Creative Constraints

Any modular synthesiser has a limited range and number of modules, this is in contrast to working in software, where aside from any CPU overhead, there is an unlimited number of devices and range of devices[2]. This limitation forces the modular synthesiser composer to problem solve, working with and around limits, compromising and innovating with what is available, and using these constraints to drive creativity (Stokes 2005).

4.2. Limited Iterations and Modules

The limit of available modules to a given synthesiser instrument also has a practical implication for the generative systems that are devised; complex interconnection and greater reliance between devices is encouraged. For example, a low frequency oscillator may have multiple waveform outputs, each affecting a different aspect of the composition, in this case any change to the oscillator frequency will impact the structure in a range of ways.

4.3. Functionality Across Broad Timescales

The continuous scale from audible frequency to compositional structure which the modular synthesiser operates in, foregrounds the underlying function of the module devices as they must be considered in the two contexts of sound and event generation simultaneously. This encourages the composer to think laterally about the possibilities inherent in a device or a configuration of multiple devices. For example, a consideration of the impact that modulating a variable pulse width waveform oscillating at low frequency requires an understanding of the underlying function beyond the audible phenomena of an oscillating pulse waveform.

4.4. Overlapping Embedded Design Philosophies

Modular synthesiser modules function at a higher level than the individual object devices made available in a software context such as MAX or Pure Data, necessarily carrying a greater level of design intention, philosophy and aesthetic preferences. Functioning on a scale somewhere in between a self contained instrument and its separated constituent components, the modular synthesiser musician is operating in a field of overlapping, interconnected (semi) instrument designs. In response to the surge in modular synthesis popularity both Cycling 74 and Native Instruments have developed standardised systems for MAX and Reaktor respectively that enable the design and use of higher level devices that emulate the

[2] Comparisons are often made between between modular synthesisers and patching software such as MAX, Pure Data or Reaktor, the points of similarity are clear in the connecting of signals between discrete objects and modularity of objects. It is unclear which technology established the modular approach first, the computer music pioneer; Max Mathews suggested that the approach was simultaneously developed by himself in the field of computer music and others working with analog electronics (Roads and Mathews 1980, 16). The motivation behind the design decision for Mathews was to create an instrument building environment that did not impose his tastes or ideas upon the musician (Roads & Mathews, 1980, p. 16), contemporary visual programming languages have continued on this trajectory offering an open environment where almost anything seems possible.

modular synthesiser paradigm ("BEAP - Powerful Modules for Max for Live," 2013) ("Reaktor 6 Blocks are like getting a modular in your laptop for $199," 2015).

4.5. Continuous Signal

In learning to use a modular synthesiser after working in a computer based compositional environment, the first conceptual hurdle is often in relation to Voltage Controlled Amplifiers (VCA) and the nature of electrical signals in the modular environment. Modular synthesisers invert the digital and keyboard centred paradigm of the note on, but also broader acoustic instrument paradigms; the plucking or bowing of a string, the striking of a drum and the blowing of a reed or brass instrument, because the electrical signal is always present while the instrument is switched on. A VCA is employed in a modular system not to produce sound, but to prevent it and attenuate it, the modulation of the VCA with an envelope or other control voltage source, allows for a signal to pass through (H. S. Howe 1975). A module must still be patched up in order for its signal to be audible or to have an effect upon other modules, but once a connection is made the signal is continuous. The break in the cause and effect relationship of the musician to the instrument may in turn impact upon the process of composition with a modular synthesiser, where signal flows are connected, attenuated and muted in order to create musical form and structure. Composer and sound designer Suzanne Chiani's description of her experience of the Buchla instruments resonates; "It's a living thing - you feel like it's alive when you're interacting with it" (Friedlander 2017).

5. CONCLUSION AND NEXT STEPS

Generative composition was a key aspect of synthesisers since the development of commercial models in the mid 1960s. This aspect of synthesis was set aside in the evolution of commercial, fixed architecture synthesisers in the early 1970s but continued to be developed through modular synthesisers. The resurgence of interest and commercial viability of modular synthesisers has resurfaced these ideas and made them available to contemporary electronic and experimental music composers.

Further scholarly attention to modular synthesisers in general is needed, the contemporary resurgence of these instruments has opened up a range of possibilities and inquiries that may re-frame experimental and electronic music practices. The research presented in this paper is a first step in exploring the nature and significance of generative compositional approaches enabled by modular synthesisers. Further research is needed to understand the methods and approaches employed by composers and their experiences of the instruments as compositional devices. This should be supported by an in-depth study of key historical and contemporary instruments, and the compositional affordances of significant module designs and configurations. This work may serve to foreground the rich history of analog systems in the compositional domain and to stimulate further creative research in the area of generative composition with modular systems.

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